Enteric Viral Pathogens in Infants in the United States: 2017–2020

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abstract

OBJECTIVE: The introduction of rotavirus vaccines in the United States led to substantial reductions in rotavirus disease burden, whereas the relative burden of other viral pathogens causing acute gastroenteritis (AGE) increased. Our aim was to establish the community-level burden of enteric viral pathogens in a US cohort during the first 2 years of life.

METHODS: From birth until age 2 years, weekly stools and AGE symptom surveys were collected from mother-child pairs in the PREVAIL (Pediatric Respiratory & Enteric Virus Acquisition and Immunogenesis Longitudinal) cohort. Stools were tested for 5 enteric viral pathogens, with infections defined by grouping proximal viral detections with the same genotype. Rates of infection and AGE symptoms were determined.

RESULTS: From 2017 to 2020, 245 children were followed, with 13 944 stools tested and 836 episodes of AGE recorded. Children had 3.8 enteric viral infections per child-year and over two-thirds had 1 or more symptomatic enteric viral infection by age 2 years. Norovirus Genogroup II (0.4 per child-year) and sapovirus (0.2 per child-year) had the highest incidence of symptomatic infections, followed by astrovirus, rotavirus, norovirus Genogroup I, and adenovirus 40/41 (each ≤0.1 per child-year). Approximately two-thirds of enteric viral infections were asymptomatic, and 90% were not medically attended. Symptomatic infections were infrequent before age 6 months.

CONCLUSION: Our findings show the high burden of symptomatic enteric viral infections, particularly norovirus, as well as the high frequency of asymptomatic infections, thereby improving our understanding of the epidemiology and clinical manifestations of enteric viral infections in young US children.

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Dr Cannon performed formal analysis, performed software programming, performed data visualization, drafted the initial and revised manuscript, and performed data curation. Dr Baker performed formal analysis, performed software programming, critically reviewed and revised the manuscript, and performed data curation. Ms Mattison performed formal analysis, critically reviewed and revised the manuscript, and performed data curation. Ms Browne, Mr Nguyen, Mr Bartlett, Ms Embury, Ms Mijatovic-Rustempasic, and Dr Casey-Moore performed laboratory analyses and results interpretation. Dr Burke performed software programming, performed data curation, and critically reviewed the manuscript. Dr Gautam provided oversight, coordinated and supervised data collection, and critically reviewed the manuscript. (Continued)

WHAT'S KNOWN ON THIS SUBJECT: In countries with high rotavirus vaccine effectiveness, the relative burden of norovirus, sapovirus, and other viral pathogens causing acute gastroenteritis (AGE) increased. However, the community-level burden of these pathogens in young US children has not been fully characterized.

WHAT THIS STUDY ADDS: By age 2, nearly all children in this US birth cohort had an enteric viral infection. Norovirus Genogroup II and sapovirus were high-burden AGE pathogens; however, asymptomatic infections were frequently detected, especially before age 6 months.

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INTRODUCTION

In children aged younger than 5 years, acute gastroenteritis (AGE) is a major cause of morbidity and mortality worldwide.[1] As of 2024, rotavirus continues to be the leading cause of AGE globally.[2] In the United States, the introduction of rotavirus vaccines (Rotarix and RotaTeq) contributed to an 80% rate reduction in rotavirus hospitalizations compared with pre-vaccine licensure.[3] Simultaneously, the relative burden of norovirus and other enteric viral pathogens has been amplified.[4–6] Norovirus alone causes an estimated 4.2 million cases of AGE, 815 000 outpatient visits, 130 000 emergency department visits, and 24 600 hospitalizations in children annually, of which children aged younger than 5 years have the highest disease burden.[7] Improved diagnostic testing capabilities have increased detection of other enteric viral pathogens (sapovirus, astrovirus, and enteric adenoviruses 40/41), raising questions about their contributions to overall AGE disease burden.[6,8–11]

Birth cohort studies have substantially enhanced our understanding of disease burden and natural history of enteric viral infections.[12] A Mexican birth cohort study established that natural infections conferred protection against rotavirus gastroenteritis, forming the basis for the development of currently licensed rotavirus vaccines.[13] Birth cohort studies conducted in low- and middle-income countries suggest that established enteric viral pathogens, such as norovirus, sapovirus, astrovirus, rotavirus, and adenovirus 40/41 are among the top pathogens causing diarrhea in young children.[9] Whether these viral pathogens equally contribute to the AGE burden in high-income settings is uncertain. A recent birth cohort study in Australia found nearly all enrolled children had at least 1 enteric viral infection by age 2 years.[14] However, no US birth cohort study focusing on AGE has been conducted. The Pediatric Respiratory and Enteric Virus Acquisition and Immunogenesis Longitudinal (PREVAIL) birth cohort study was designed to determine the natural history of enteric and respiratory viral infections and immune response to those infections in US children.[15] Here, we describe the burden of enteric viral infections among children in the PREVAIL cohort.

METHODS

Mother-child pairs residing in Cincinnati, Ohio, were enrolled into the PREVAIL cohort from April 2017 to July 2018. Children were followed from birth until age 2 years, with data collection ending in July 2020. The study design, enrollment strategy, inclusion criteria, and methodologies for data and sample collection were previously described.[15] Information on childcare (eg, location, days per week) and breastfeeding duration was collected at age 6 weeks and 4, 6, 12, 18, and 24 months. The study was approved by the institutional review boards at the Centers for Disease Control and Prevention (CDC), Cincinnati Children's Hospital Medical Center, and the enrolling birth hospitals.

Stool Collection and Symptom Definitions

Stool specimens from children were collected by mothers on a weekly basis; additional stools were requested when the child was reported to have AGE symptoms. AGE was defined as 3 or more instances of diarrhea or 1 or more episodes of vomiting within a 24-hour period. Symptoms, medical care use, and start and end dates were reported by mothers in text surveys or by medical record abstraction.

Stool Testing Workflows

Nucleic acids from stool specimens were extracted using commercial kits for viral RNA, and reverse transcriptase-polymerase chain reaction (RT-PCR) was performed to detect norovirus Genogroup I (norovirus) (GI), norovirus Genogroup II (norovirus) GII, sapovirus, astrovirus, and rotavirus.[8,16,17] The RT-PCR testing workflow and additional testing for enteric adenovirus 40/41, 9 enteric bacteria and 3 protozoa[18] by the Luminex xTAG Gastrointestinal Pathogens Panel is described in the Supplemental Methods and Supplemental Figure 1. A pathogen detection was defined by presence of its nucleic acid in a stool by any detection method. Stools that tested positive for norovirus GI, norovirus GII, sapovirus, astrovirus, and rotavirus were genotyped.[8,16,17] Rotavirus vaccine strain shedding was excluded; only wild-type rotavirus detections were considered.

Viral Infection Definitions

Enteric viral infections were defined by grouping viral detections of the same genotype occurring within 30 days of the previous detection (Supplemental Figure 2). A viral infection was classified as symptomatic if the first positive stool was collected 4 days or fewer before or 7 days after symptoms began and asymptomatic if no symptoms were reported from 4 days before through 14 days after the first positive stool. Symptom status was considered unknown if symptom surveys were incomplete or symptoms were already assigned to a previous, different viral infection (Supplemental Figure 3).

Statistical Analysis

We used data from the full cohort except when calculating the proportion of children having at least 1 viral infection by ages 6 months, 1 year, and 2 years and the duration of viral shedding in stools. For these calculations, a subset of participants considered to be adherent, who submitted at least 70% of weekly child specimens and completed at least 18 months of follow-up, was used because consistent follow-up would have a significant impact on these calculations. We excluded co-infections (infection by more than 1 viral pathogen beginning on the same day or a nonviral

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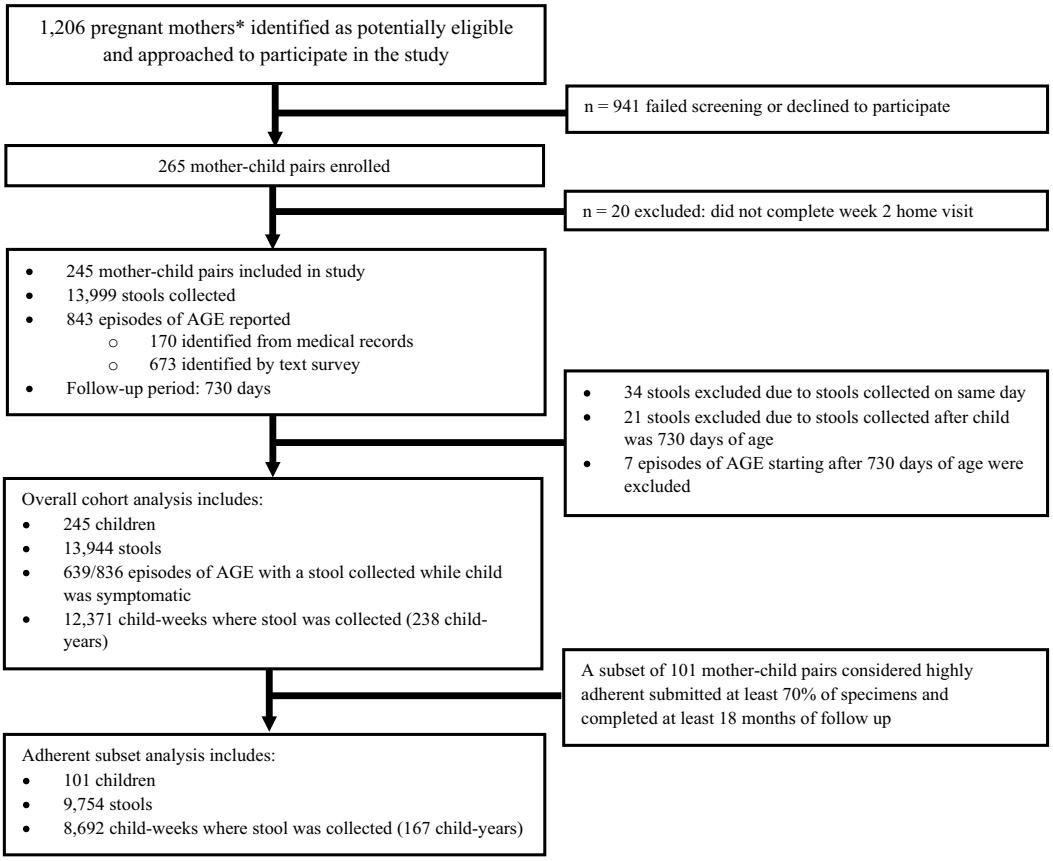
pathogen detected during symptoms) to ensure our findings were specific for viral pathogens.

The χ^2 test or Fisher exact test was used for comparisons of stool prevalence and proportional differences between subgroups. The Wilcoxon rank sum test was used for comparing median durations of viral shedding and for comparing the ages at which children experienced their first asymptomatic vs symptomatic infections. The incidence rates (IRs) of viral infections were calculated using the number of weeks in which a stool specimen was submitted per participant as the denominator (weeks at risk) without subtraction because children continue to be at risk of overlapping infections by different enteric viral pathogens or different genotypes of the same viral pathogen. All statistical analysis was performed using the R Environment for Statistical Computing version 4.4.0 (R Foundation for Statistical Computing, Vienna, Austria).

RESULTS

Characteristics of the Overall Cohort and Adherent Subset

Of the enrolled mother-child pairs, 245 met the eligibility criteria for the study (Figure 1). Demographic and cohort characteristics of eligible and enrolled participants were described previously,[15,19] with differences in the demographic and socioeconomic characteristics of the mother-child dyads in the adherent (n = 101; 41%) and less-adherent (n = 144; 59%) subsets, as well as differences in breastfeeding initiation and duration (Supplemental Table 1). Rotavirus vaccine was received by 92% of children, with similar vaccination rates among adherent (95%) and less-adherent subsets (90%) (Supplemental Table 1). Adherent mothers submitted 70% of all stools (9,754 of 13,944) and 80% (4,791 of 5,990) of stools collected during the second



* Based on obstetrical records, potentially eligible pregnant mothers identified as 44.78% White (n=540), 47.60% Black (n=574), and 7.63% of other or unknown (n=92) race.

FIGURE 1.
Flowchart of study participation, specimen collection, and survey completion among mother-child pairs enrolled in the PREVAIL cohort, April 2017 through July 2020.
Abbreviation: PREVAIL, Pediatric Respiratory & Enteric Virus Acquisition and Immunogenesis Longitudinal.

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TABLE 1. Number of Enteric Viral Infections, Incidence of Infections and Symptomatic Enteric Viral Infections, and Proportion Symptomatic, for Children in the PREVAIL Cohort (n = 245) During the First 2 Years of Life, April 2017 Through July 2020

Viral Pathogen	Number of Viral Infections, N[a]	Viral Infections per Child-Year, IR (95% CI)[a]	Symptomatic Viral Infections, n (%)	Symptomatic Viral Infections per Child-Year, IR (95% CI)
Norovirus GII	301	1.3 (1.1–1.4)	98 (33)	0.4 (0.3–0.5)
Sapovirus	224	0.9 (0.8–1.1)	46 (21)	0.2 (0.1–0.3)
Astrovirus	122	0.5 (0.4–0.6)	20 (16)	0.1 (0.1–0.1)
Rotavirus	102	0.4 (0.4–0.5)	21 (21)	0.1 (0.1–0.1)
Norovirus GI	67	0.3 (0.2–0.4)	16 (24)	0.1 (0.0–0.1)
Adenovirus 40/41[b]	47	0.3 (0.3 – 0.5)	8 (17)	0.1 (0.0 – 0.1)
Viral co-infection	41	0.2 (0.1–0.2)	13 (32)	0.1 (0.0–0.1)
Any viral infection	904	3.8 (3.6–4.1)	222 (25)	0.9 (0.8–1.1)

Abbreviations: IR, incidence rate; GI, Genogroup I (norovirus); GII, Genogroup II (norovirus); PREVAIL, Pediatric Respiratory & Enteric Virus Acquisition and Immunogenesis Longitudinal.

[a] Viral infections with an unknown symptom status (n = 112) are included in total number of infection values and in IR calculations.

[b] Adenovirus 40/41 testing was performed only on stools collected during the first year of life.

year of life (Supplemental Figure 4). Adherent mothers completed symptom surveys for a median of 99% (IQR 96–100%) of weeks they were active in the study, as opposed to a median of 73% (IQR 57–91%) for the less-adherent subset. Overall, of the 13 944 stools tested, 91% (n = 12 697) had symptom surveys or medical records available.

Incidence of Viral Infections and Proportion of Children Infected

A total of 904 enteric viral infections were identified among children in the overall cohort: 25% symptomatic (222 of 904), 63% asymptomatic (570 of 904) and 12% (112 of 904) with an unknown symptom status. Nine of 10 (90%) children in the overall cohort experienced at least 1 enteric viral infection. Viral infection and symptomatic viral infection IRs for the overall cohort were 3.8 and 0.9 per child-year, respectively (Table 1). A total of 97% (98 of 101) of children in the adherent subset had 1 or more enteric viral infections, and 71% (72 of 101) of adherent children had at least 1 symptomatic enteric viral infection by age 2 (Figure 2 and Supplemental Figure 5). A higher proportion of children in the adherent subset had a symptomatic enteric viral infection by age 2 years when compared with the overall cohort (54%) (Supplemental Figure 5).

Norovirus GII and sapovirus were the most common infections identified (Table 1). The IR of norovirus GII was 1.3 infections and 0.4 symptomatic infections per child-year for the overall cohort (Table 1). A total of 80% of adherent children (81 of 101) had at least 1 infection and half (50 of 101) had a symptomatic infection by age 2 (Figure 2). The IR for sapovirus was 0.9 infections and 0.2 symptomatic infections per child-year for the overall cohort (Table 1). Three fourths (74%) adherent children (75 of 101) had 1 or more infection, and 30% (30 of 101) had a symptomatic infection by age 2 (Figure 2). For all other enteric viral pathogens among the overall cohort, individual IRs were ≤0.5 infections and ≤0.1 symptomatic infections per child-year by age 2 (Table 1).

Viral Infections in the First Year of Life

By age 1, approximately 80% of adherent children had at least 1 enteric viral infection and one-third had a symptomatic enteric viral infection (Figure 2). Among infants aged 0 to 5 months in the overall cohort, IRs for enteric viral infection (2.4 per child-year) and symptomatic infection (0.4 per child-year) were lower than those among infants aged 6 to 11 months (4.9 and 1.2 per child-year, respectively) (Supplemental Table 2). This pattern was observed for norovirus GII, sapovirus, astrovirus and norovirus GI, but not for rotavirus and adenovirus 40/41 (Supplemental Table 2). First asymptomatic infections occurred at a younger median age than first symptomatic infections for norovirus GII ($P < .001$), sapovirus ($P < .01$), and adenovirus 40/41 ($P < .01$); for rotavirus ($P = .22$), norovirus GI ($P = .31$), and astrovirus ($P = .72$), these differences were not significant (Figure 3).

Symptomology and Medical Care Use for Enteric Viral Infections

Overall, 64% (548 of 863) of single-pathogen enteric viral infections among children in the overall cohort were asymptomatic, 24% (209 of 863) were treated by homecare, and 10% were medically attended (Figure 4A). Among symptomatic enteric viral infections in the overall cohort, approximately one-third were medically attended (35%; 73 of 209); 37% for norovirus GII, 28% for sapovirus, 30% for astrovirus, 43% for rotavirus, 31% for norovirus GI, and 50% for adenovirus) (Figure 4B). There were 24 emergency department visits (15 for norovirus GII, 2 each for sapovirus, astrovirus, rotavirus and norovirus GI, and 1 for adenovirus), as well as 3 hospitalizations (2 associated

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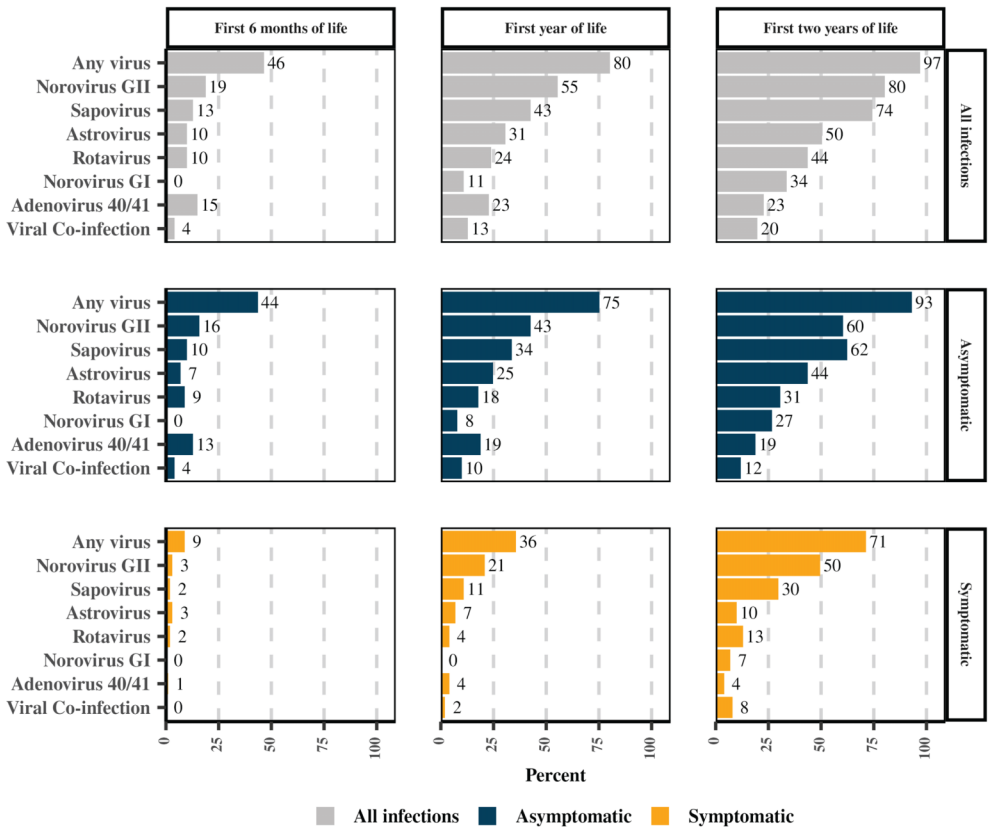

*Viral infections with an unknown symptom status (n=47) were included in the "all infections" category. Adenovirus 40/41 testing was performed only on stools collected during the first year of life.

FIGURE 2.
Percentage of adherent children (n = 101) having at least 1 enteric viral infection during the first 2 years of life in the PREVAIL cohort, April 2017 through July 2020, by symptom status and age category.
Abbreviation: PREVAIL, Pediatric Respiratory & Enteric Virus Acquisition and Immunogenesis Longitudinal.

with norovirus GII and 1 associated with rotavirus). Symptoms varied by pathogen, but overall, most children experienced both vomiting and diarrhea during AGE episodes (52%, 108 of 209), with 30% (62 of 209) reporting diarrhea only and 18% (38 of 209) reporting vomiting only (Figure 5).

AGE Episodes and Medically Attended AGE

Children in the overall cohort experienced 836 episodes of AGE by age 2, with 84% (205 of 245) of children having at least 1 episode. The median number of AGE episodes observed per child was 3.0 (IQR 1.0–5.0). At least 1 stool was collected during symptoms or within 7 days after for 76% (639 of 836) of episodes. Enteric viral pathogens were detected in stools from 242 of 639 (38%) AGE episodes (Figure 6), among which, norovirus GII (115 of 639; 18%) was most often detected (Figure 6). Among medically attended episodes, enteric viral pathogens were detected in 82 of 244 (34%), most commonly norovirus GII (16%; 39 of 244), with other or mixed enteric viral pathogens detected in ≤5% (Figure 6). Testing stools from a subset of AGE episodes showed nonviral enteric pathogens present as single detections in 3% (15 of 491) of AGE episodes ("Supplemental Results" section and Supplemental Table 3).

Viral Shedding in Children From Whom Stools Were Collected per Protocol

During single-pathogen infections, enteric viral pathogens were shed in adherent stools of children (n = 101) for a median of 6.0 days [IQR 1.0–16.0], with norovirus GII shed longer than other viral pathogens (sapovirus $P < .001$; astrovirus $P < .001$; rotavirus $P < .01$; adenovirus 40/41 $P < .01$) except norovirus GI ($P = .02$) (Supplemental Table 4). Children shed norovirus GII longer during symptomatic infections (median 18 days [IQR 10.0–29.0]) than in asymptomatic infections (median 12.0 days [IQR 1.0–22.0]) ($P < .01$) (Supplemental Table 4), which is a pattern shared for rotavirus and sapovirus infections but not for the other viral pathogens (Supplemental Table 4).

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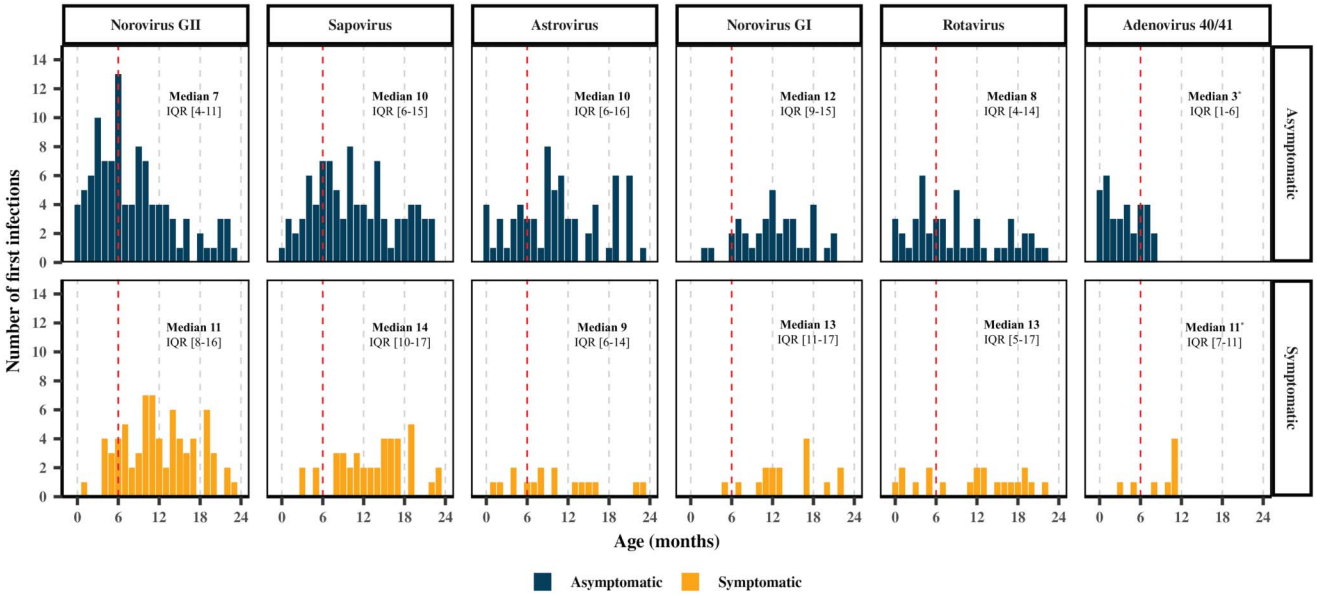
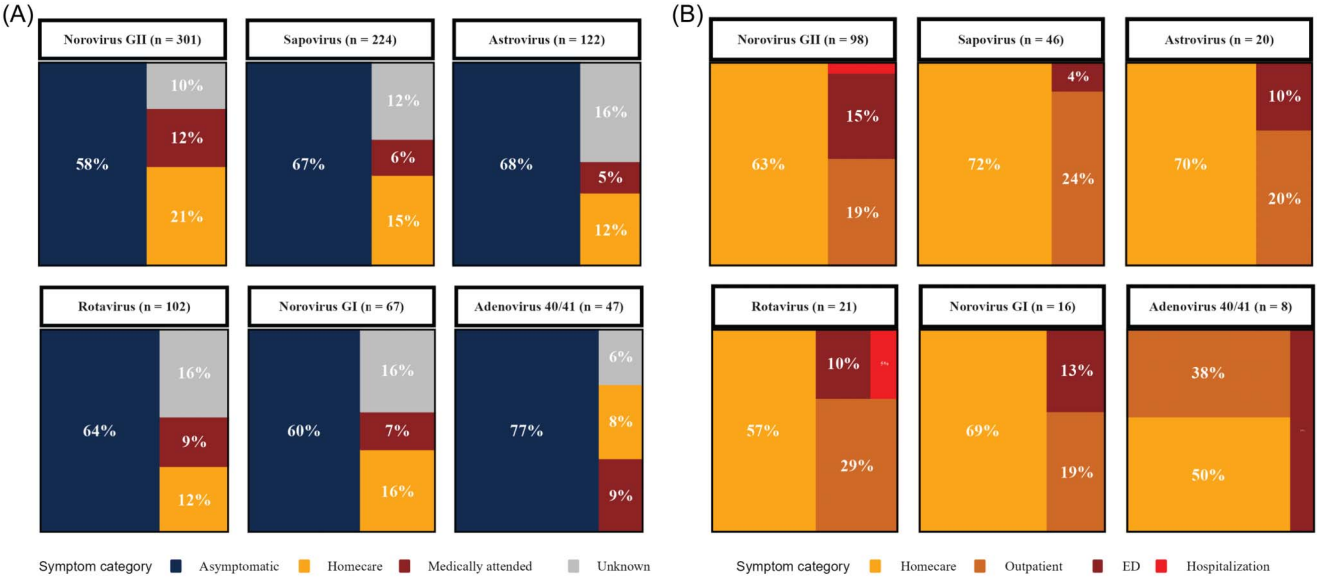


*Adenovirus 40/41 testing was only performed on stools collected during the first year of life; interquartile range (IQR); viral infections where more than one enteric pathogen was deteced were excluded from analysis to ensure our findings were specific for each viral pathogen.

FIGURE 3.
Age (months) at which first asymptomatic and symptomatic enteric viral infections occurred in children in the PREVAIL cohort (n = 245) during the first 2 years of life, April 2017 through July 2020.
Abbreviation: PREVAIL, Pediatric Respiratory & Enteric Virus Acquisition and Immunogenesis Longitudinal.



*Viral infections where more than one enteric pathogen were detected were excluded from analysis to ensure our findings were specific for each viral pathogen; Adenovirus 40/41 testing was only performed on stools collected during the first year of life.
**Percent values in panels total 99% or 101% in some cases due to rounding.
***Hospitalizations accounted for 2% of symptomatic norovirus GII infections and 5% of rotavirus symptomatic infections. Emergency department (ED) visits accounted for 15% of adenovirus 40/41 symptomatic infections.

FIGURE 4.
Symptomology and medical care attendance for single-pathogen enteric viral infections* (n = 863) among children in the PREVAIL cohort (n = 245) during the first 2 years of life, April 2017 through July 2020. Categories of symptomology and medical attendance by (A) all enteric viral infections** and (B) symptomatic enteric viral infections***.
Abbreviation: PREVAIL, Pediatric Respiratory & Enteric Virus Acquisition and Immunogenesis Longitudinal.

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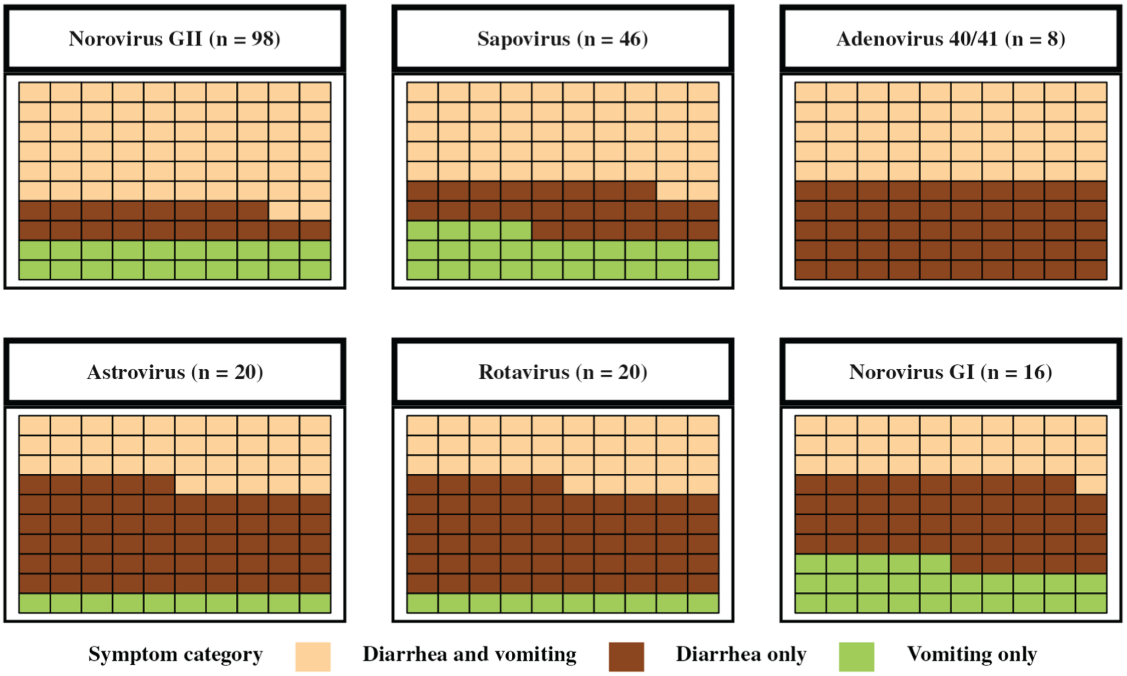
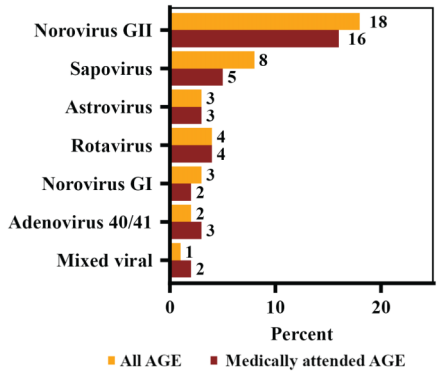


*Viral infections where more than one enteric pathogen was detected were excluded from analysis to ensure our findings were specific for each viral pathogen; Adenovirus 40/41 testing was only performed on stools collected during the first year of life.

FIGURE 5.
Proportions of symptomatic enteric viral infections characterized by symptoms of diarrhea and vomiting, diarrhea only, and vomiting only among children in the PREVAIL cohort (n = 245), April 2017 through July 2020.
Abbreviation: PREVAIL, Pediatric Respiratory & Enteric Virus Acquisition and Immunogenesis Longitudinal.

Seasonality of Enteric Viral Infections

Norovirus GII showed a winter seasonality with infections among children in the overall cohort peaking between November and March (Supplemental Figure 6). Sapovirus followed a similar pattern during the 2018 to 2019 season, but the peak shifted toward late fall to early winter for the 2019 to 2020 season (Supplemental Figure 6). Although fewer in number, astrovirus, rotavirus, and norovirus GI infections were detected primarily in the late winter and early spring, and adenovirus 40/41 infections had no apparent seasonality (Supplemental Figure 6).



*Adenovirus 40/41 testing was only performed on stools collected during the first year of life (n = 345 AGE episodes and n = 159 medically-attended AGE episodes). AGE episodes with more than one viral pathogen detected were considered "mixed viral".

FIGURE 6.
Enteric viral pathogen detection in stools from AGE episodes and medically attended AGE episodes among children (n = 245) during the first 2 years of life in the PREVAIL cohort, April 2017 through July 2020.
Abbreviation: PREVAIL, Pediatric Respiratory & Enteric Virus Acquisition and Immunogenesis Longitudinal.

DISCUSSION

The PREVAIL study followed 245 mother-child pairs who provided nearly 14 000 stool specimens, 90% of which linked to symptom surveys and medical records, providing a comprehensive assessment of the epidemiology and clinical manifestations of enteric viral infections in a US population of children aged younger than 2 years. Nearly all adherent children experienced at least 1 enteric viral infection, with two-thirds having at least 1 symptomatic enteric viral infection by age 2 years. Children had 3.8 enteric viral infections per child-year, including 0.9 symptomatic viral infections per child-year. By age 2, 80% of adherent children had at least 1 norovirus GII infection, 74% had at least 1 sapovirus infection, and half had at least 1 infection by astrovirus, rotavirus, norovirus GI or adenovirus 40/41. Despite the high incidence of symptomatic infections, most enteric viral infections were asymptomatic, and 10% were

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medically attended, underscoring the pervasiveness of mild-to-moderate and subclinical infections in young children.

Norovirus GII and sapovirus were the most common enteric viral infections in early life, consistent with findings from an Australian birth cohort[14,20,21] and in other countries.[9,12,22–25] Norovirus (GI or GII) was associated with 21% of AGE episodes and 18% of medically attended AGE episodes, similar to the 23% reported in a US population-based study of children aged over 2 years.[26] In that study, sapovirus, rotavirus, and astrovirus were detected in 7%, 3%, and 3% of medically attended AGE,[26] compared with 5%, 4%, and 3% among PREVAIL children. This cohort had high rotavirus vaccination coverage (92%) and previously demonstrated robust, albeit complex, immune responses to rotavirus vaccination.[17] Only 13% of adherent children had at least 1 symptomatic rotavirus infections by age 2, with 9% of all rotavirus infections medically attended. Ongoing PREVAIL studies are assessing longitudinal responses to rotavirus vaccination, wild-type infection, and host-specific mother-child factors influencing immunity.

Few children had symptomatic enteric viral infections during their first 6 months of life. Symptomatic norovirus GII and sapovirus infection incidence rose sharply thereafter, suggesting early protection followed by waning of maternal antibodies and maternal immune cells coinciding with more frequent infant interaction with the environment. Assessing the influence of maternal antibody transfer on infant infection risk and response following natural infection or vaccination[17] is important for optimizing timing for vaccine delivery.[17,27] A norovirus vaccine candidate administered to infants aged around 5 months underperformed at study endpoints,[28] perhaps related to maternal antibody interference or imprinting by an unrelated first infection. Advancing pediatric vaccine development has significant public health implications given the demonstrated success of rotavirus vaccines and the considerable burden of norovirus AGE in children. PREVAIL studies investigating norovirus-specific maternal and child antibodies, including neutralizing antibodies, are underway.

A strength of our study is its high frequency of stool collection and testing, overcoming limitations of birth cohort studies with less frequent stool testing[9,12,22–25,29,30] and studies that rely only on medically attended cases,[10,31] which would underestimate mild illnesses and asymptomatic infections. Sustained circulation of viral pathogens through asymptomatic carriage creates potential for viral transmission to vulnerable populations and opportunities for emergence and spread of new viral strains.[32] Asymptomatic infections and post-symptomatic shedding of enteric viral pathogens have led to outbreaks of norovirus and household transmission.[33,34] Additionally, asymptomatic or mild infections may prime the immune system to prevent more severe illnesses following exposure, a hypothesis serological studies will address in future PREVAIL studies.

Our study has several limitations. Mothers in the adherent and less-adherent subgroups differed in characteristics that limit the generalizability of our findings related to exposures and AGE acquisition, including social determinants of health (eg, education, socioeconomic status, single vs partnered household, race) and breastfeeding initiation and duration. We used data from the full cohort except when consistency of follow-up would have significant impact on findings, yet recognize the need for studies that better reflect the broader population. Exact infection start and end dates could not be determined, which impacted the duration of stool shedding findings. COVID-19 mitigation could have impacted specimen submission and reporting, although study children completed at least 20 months of follow-up by early March 2020. The multipathogen panel used for adenovirus 40/41 testing made it cost-prohibitive to test all stools in the second year of life, particularly because testing a subset of stools from AGE episodes showed low prevalence of bacterial and protozoan pathogens. Multiplex quantitative reverse transcriptase-polymerase chain reaction is a lower-cost alternative for which a duplex norovirus GI/GII assay is available, and we are currently developing a test for the simultaneous detection of adenovirus 40/41, sapovirus, and astrovirus. Together, these will allow for detection of around 80% of PREVAIL enteric viral infections.

CONCLUSION

Our study enhances our understanding of epidemiology, symptoms, and disease burden of enteric viral pathogens, providing a broader perspective on the range of symptoms in early childhood. Symptomatic viral AGE infections, especially those caused by norovirus GII and sapovirus, were frequent, underscoring the need for pediatric norovirus vaccines and drawing attention to the importance of sapovirus. Asymptomatic infections were even more common, but the extent of their role in transmission is unclear. Our findings provide an extensive overview of the burden of enteric viral infections in young children, a foundation for a more complete understanding of AGE epidemiology, infection, and immunity.

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ABBREVIATIONS

AGE: acute gastroenteritis
CDC: Centers for Disease Control and Prevention
GI: Genogroup I (norovirus)
GII: Genogroup II (norovirus)
IR: incidence rate
PCR: polymerase chain reaction
PREVAIL: Pediatric Respiratory & Enteric Virus Acquisition and Immunogenesis Longitudinal
RT-PCR: reverse transcriptase-polymerase chain reaction

Ms Burrell and Dr Conrey critically reviewed the manuscript and performed data curation. Dr Parashar provided oversight and critically reviewed the manuscript. Dr Payne conceptualized and designed the study, provided oversight, coordinated and supervised data collection, and critically reviewed the manuscript. Dr Staat conceptualized and designed the study, provided oversight, coordinated and supervised data collection, and critically reviewed the manuscript. Dr Morrow conceptualized and designed the study, provided oversight, coordinated and supervised data collection, and critically reviewed the manuscript. Dr Vinjé conceptualized and designed the study, provided oversight, coordinated and supervised data collection, and critically reviewed the manuscript. All authors approved the final manuscript as submitted and agree to be accountable for all aspects of the work.

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